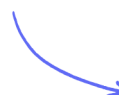


Structured Questions

Organic Chemistry: Halogenoalkanes

Halogenoalkanes - Introduction / Reactions of Halogenoalkanes / The Nucleophilic Substitution Mechanism / Hydrolysis of Halogenoalkanes / Trends in Halogenoalkanes

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Total Marks

/43

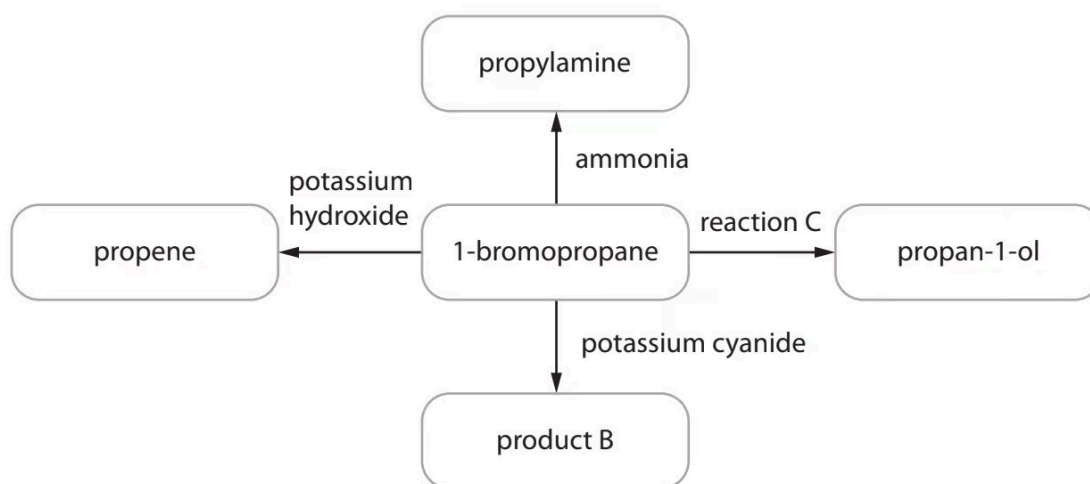
1 (a) 1-bromopropane is used for spot removal when 'dry cleaning' clothes.

1-bromopropane does not occur naturally but can be made from propan-1-ol.

Identify the reagent or reagents you would use to make 1-bromopropane from propan-1-ol.

(1 mark)

(b) Some reactions of 1-bromopropane are shown.



i) Give the conditions for the formation of propene.

(1)

ii) Give the **molecular** formula of product B.

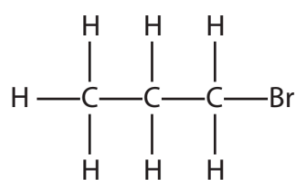
(1)

iii) Name the type and mechanism of the reaction taking place in reaction C.

(2)

iv) Complete the mechanism for the reaction that occurs between ammonia and 1-bromopropane to form propylamine, $\text{CH}_3\text{CH}_2\text{CH}_2\text{NH}_2$. Include curly arrows, and relevant lone pairs and dipoles.

(4)



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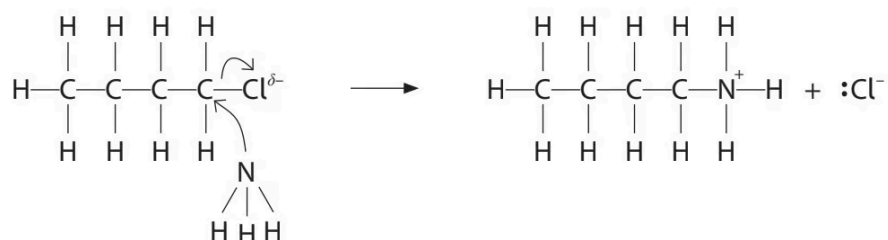
(8 marks)

2 (a) Ammonia reacts with 1-chlorobutane.

State the type and mechanism of this reaction.

(2 marks)

(b) A student drew the first step of the mechanism for the reaction.

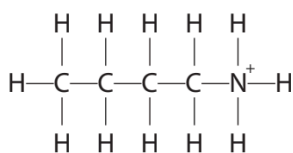


i) Identify **two** omissions in the student's mechanism.

(2)

ii) To obtain butylamine, sodium hydroxide solution is added. Complete the next step of the mechanism to form butylamine, showing curly arrows, relevant lone pairs and the reaction products.

(3)



(5 marks)

- (c)** The reactions of ammonia and of hydroxide ions with halogenoalkanes are similar.

Compare the rate of reaction of ammonia with 1-chlorobutane and with 2-bromo-2-methylpropane. Justify your answer.

(3 marks)

3 (a) This question is about halogenoalkanes.

The rates of hydrolysis of 1-chloropropane, 1-bromopropane and 1-iodopropane in reactions with aqueous silver nitrate solution were compared.

i) State what would be measured in the experiment to compare the rates of hydrolysis.

(1)

ii) State which of these halogenoalkanes would hydrolyse the fastest. Justify your answer.

(2)

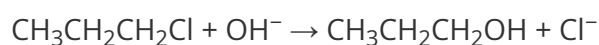
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(3 marks)

(b) The equation for the hydrolysis of 1-chloropropane by aqueous hydroxide ions is shown.



Give the mechanism for this hydrolysis. Include curly arrows, and relevant dipoles and lone pairs.

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(2 marks)

(c) The boiling temperatures of some halogenoalkanes are shown.

Halogenoalkane	Boiling temperature / °C
1-chloropropane	47
1-bromopropane	71
1-iodopropane	103

Explain the trend in boiling temperature of these halogenoalkanes by comparing the intermolecular forces involved.

Detailed explanations of the forces involved are not required.

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(4 marks)

- (d) In another experiment, 2-bromobutane is heated with ethanolic potassium hydroxide and an elimination reaction occurs.

Draw the **skeletal** formulae of the three possible organic products, giving their names.

Skeletal formula	Name

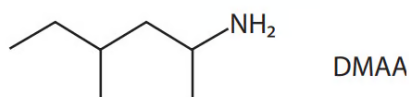
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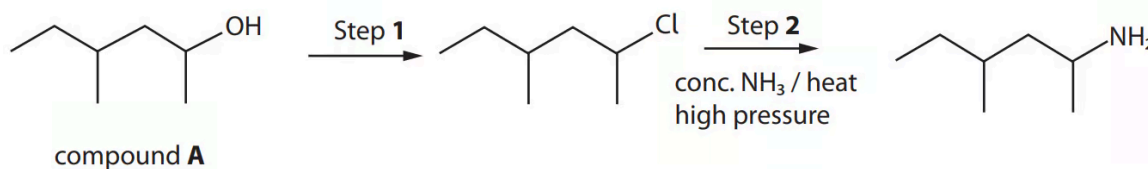
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(3 marks)

4 (a) The compound DMAA was originally synthesised as a decongestant.



A suggested synthetic route for DMAA is shown.



i) Give the systematic name of compound **A**.

(1)

ii) Identify, by name or formula, a suitable reagent for Step 1.

(1)

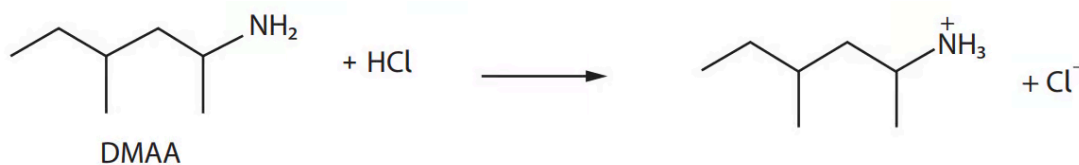
iii) Give the mechanism for the reaction in Step 2.

Include curly arrows, and any relevant dipoles and lone pairs.

(4)

(6 marks)

- (b) DMAA is only slightly soluble in water but dissolves readily in hydrochloric acid to form an aqueous solution.

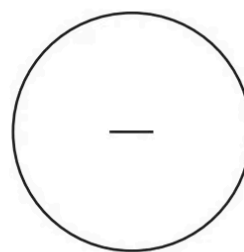
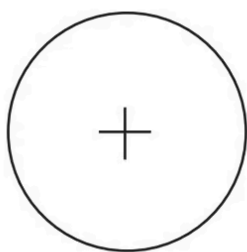


i) Explain the type of bonding that occurs between the nitrogen atom and the hydrogen ion, when the positive ion forms.

(3)

ii) Complete the diagrams to show how the ions formed in the reaction between DMAA and hydrochloric acid interact with water molecules.

(3)



(6 marks)